

Monday

1. Test 2 is Next Friday
 - A. Study Guide is on coursed en
2. HW 7.2.2 7.3.2 due
 - A. Questions? (number 4)
 - B. Questions from 7.3 Worksheet?
3. Pick the method I Worksheet
4. Start 7.4 Partial Fractions

$$4. \int \frac{18x^2}{\sqrt{36-x^2}} dx$$

$$\begin{array}{l} x = 6 \sin \theta \\ dx = 6 \cos \theta d\theta \\ \sqrt{36 - 36 \sin^2 \theta} \\ \sqrt{36(1 - \sin^2 \theta)} \\ \sqrt{36 \cos^2 \theta} \\ 6 \cos \theta \end{array}$$

$$\int \frac{18 \cdot 36 \sin^2 \theta \cdot \cancel{6 \cos \theta} d\theta}{\cancel{6 \cos \theta}}$$

$$= 648 \int \sin^2 \theta d\theta = 648 \cdot \frac{1}{2} \int 1 - \cos 2\theta d\theta = 324 \left[\theta - \frac{1}{2} \sin(2\theta) \right] + C$$

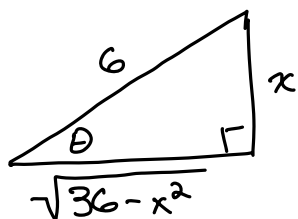
$$= 324 \left[\theta - \sin \theta \cos \theta \right] + C = 324 \left[\sin^{-1}\left(\frac{x}{6}\right) - \frac{x \sqrt{36-x^2}}{36} \right] + C$$

$2 \sin \theta \cos \theta$

$$324 \sin^{-1}\left(\frac{x}{6}\right) - 9x \sqrt{36-x^2} + C$$

$$\sin \theta = \frac{x}{6} = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{\sqrt{36-x^2}}{6}$$



$$\int \tan^3 \theta \sec^4 \theta d\theta = \int \underbrace{\tan^2 \theta}_{u^2} \underbrace{\sec^2 \theta}_{(u^2+1)} (\underbrace{\sec \theta}_{u} du) = \int \underbrace{\tan \theta}_{u} \underbrace{\sec^3 \theta}_{u^3} (\underbrace{\sec \theta \tan \theta}_{du})$$

$u = \tan \theta$
 $du = \sec^2 \theta d\theta$

OR
 $u = \sec \theta$
 $du = \sec \theta \tan \theta d\theta$

$$= \int u^2(u^2+1) du$$

$$= \int u^4 + u^2 du = \frac{1}{5} u^5 + \frac{1}{3} u^3 + C$$

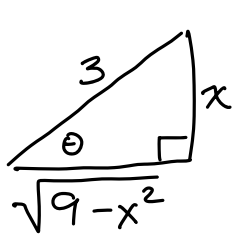
$$= \boxed{\frac{1}{5} \tan^5 \theta + \frac{1}{3} \tan^3 \theta + C}$$

$$3. \int \frac{\sqrt{9-x^2}}{5x^2} dx = \int \frac{3 \cos \theta \cdot 3 \cos \theta d\theta}{5 \cdot 9 \sin^2 \theta} = \frac{1}{5} \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta$$

$$\begin{aligned} x &= 3 \sin \theta \\ dx &= 3 \cos \theta d\theta \\ \sqrt{9-9\sin^2 \theta} &= \sqrt{9(1-\sin^2 \theta)} \\ &= 3 \cos \theta \end{aligned}$$

$$= \frac{1}{5} \int \cot^2 \theta d\theta = \frac{1}{5} \int \csc^2 \theta - 1 d\theta$$

$$= \frac{1}{5} \left[-\cot \theta - \theta \right] + C$$



$$\sin \theta = \frac{x}{3}$$

$$\theta = \arcsin\left(\frac{x}{3}\right)$$

$$\cot \theta = \frac{\text{adj}}{\text{opp}}$$

$$= \boxed{\frac{1}{5} \left[-\frac{\sqrt{9-x^2}}{x} - \arcsin\left(\frac{x}{3}\right) \right] + C}$$

Integration Worksheet: Pick the Method I

In groups of 2-4 people, examine each integral below and determine the best method to use on it. Give some detail about how you'd start, i.e. ...

- for IBP: tell what the u and dv should be, or if you should use the tabular method.
- for substitutions, tell what u should be or which trig substitution to use.
- for manipulations, show what to do.

1. $\int (x^2 + 3x) \sin(2x) dx$ Tabular IBP

2. $\int \frac{x^3 + 3x - 9}{\sqrt{x}} dx = \int \frac{x^3}{x^{1/2}} + \frac{3x}{x^{1/2}} - \frac{9}{x^{1/2}} dx = \int x^{5/2} + 3x^{1/2} - 9x^{-1/2} dx$

3. $\int 5x\sqrt{x^2 + 9} dx$ trig sub $\sqrt{x} = 3 \tan \theta$ u-sub $u = x^2 + 9$
 $du = 2x dx$

4. $\int \tan^2 x dx = \int \sec^2 x - 1 dx = \tan x - x + C$

5. $\int \tan^{-1}(5x) dx$ IBP
 $\arctan(5x)$

H
I
A
T
E

6. $\int \cot x \csc x dx = -\csc x + C$ $\frac{d}{dx}(\csc x) = -\csc x \cot x$

7. $\int \sin^4(x) \cos^3(x) dx$ $u = \sin x$ $\int \sin^4 x \cos^2 x (\cos x dx)$
 du

8. $\int e^x \cos(3x) dx$ IBP Twice + Solve

9. $\int \sin^4(x) dx = \int (\sin^2 x)^2 dx = \int \left[\frac{1}{2} (1 - \cos 2x) \right]^2 dx$

10. $\int \frac{\sqrt{4x^2 + 9}}{x} dx$ trig sub. $2x = 3 \tan \theta$

~~$u = 4x^2 + 9$
 $du = 8x dx$~~

~~$$\frac{\sqrt{u}}{x} \quad \frac{du}{8x}$$~~

1. tabular IBP
2. Split into $\frac{x^3}{\sqrt{x}} + \frac{3x}{\sqrt{x}} - \frac{9}{\sqrt{x}} = x^{\frac{5}{2}} + 3x^{\frac{1}{2}} - 9x^{-\frac{1}{2}}$, then use the power rule.
3. u -sub: let $u = x^2 + 9$. Trig sub $x = 3 \tan \theta$ will also work, but it's harder.
4. Rewrite $\tan^2 x$ as $\sec^2 x - 1$ and integrate.
5. IBP: $u = \tan^{-1}(5x)$ $dv = dx$
6. from the trig derivatives, this is just $-\csc x$
7. Trig integral: $\cos$ power is odd, so $u = \sin x$ $du = \cos x$
8. IBP Twice and solve for the integral.
9. Trig integral: manipulate $\sin^4(x)$ using double-angle formulas.
10. Trig Sub: let $2x = 3 \tan \theta$

7.4 Partial Fractions

In this section we will be integrating rational functions, i.e.

$$\int \frac{\text{polynomial}}{\text{polynomial}} dx.$$

Note: Before using partial fractions, you must make sure that the denominator has higher degree than the numerator. Use polynomial division if necessary.

We need to keep the following integrals in mind:

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int \frac{1}{mx+b} dx = \frac{1}{m} \ln |mx+b| + C$$

$$\int \frac{1}{x^2} dx = -\frac{1}{x} + C$$

$$\int \frac{1}{x^2 + 1} dx = \arctan(x) + C$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{x}{x^2 + a^2} dx = \frac{1}{2} \ln(x^2 + a^2) + C$$

1 Case 1: Denominator factors into distinct linear factors.

1.0.1 Example:

$$\int \frac{x}{x^2 + 3x + 2} dx = \int \frac{x}{(x+1)(x+2)} dx = \int \frac{-1}{x+1} + \frac{2}{x+2} dx$$

$$\left[\frac{x}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2} \right] (x+1)(x+2)$$

$$x = A(x+2) + B(x+1)$$

plug in things to zero out A+B.

plug in $x = -2$

$$-2 = A(-2+2) + B(-2+1)$$

$$-2 = B(-1)$$

$$\underline{2 = B}$$

plug in $x = -1$

$$-1 = A(-1+2) + B(-1+1)$$

$$-1 = A(1)$$

$$\underline{-1 = A}$$

$$\int \frac{-1}{x+1} + \frac{2}{x+2} dx = \boxed{-\ln|x+1| + 2\ln|x+2| + C}$$

1.0.2 Example:

$$\int \frac{2x+1}{x^2+2x-35} dx \quad \left[\frac{2x+1}{(x+7)(x-5)} = \frac{A}{(x+7)} + \frac{B}{(x-5)} \right] (x+7)(x-5)$$

$$2x+1 = A(x-5) + B(x+7)$$

plug in $x=5$

$$2 \cdot 5 + 1 = A(5-5) + B(5+7)$$

$$11 = B \cdot 12$$

$$\frac{11}{12} = B$$

plug in $x=-7$

$$2(-7)+1 = A(-7-5) + B(-7+7)$$

$$-13 = -12A$$

$$\frac{13}{12} = A$$

$$\int \frac{\frac{13}{12}}{x+7} + \frac{\frac{11}{12}}{x-5} dx = \boxed{\frac{13}{12} \ln|x+7| + \frac{11}{12} \ln|x-5| + C}$$

Note: $2x+1 = A(x-5) + B(x+7)$ can also be solved by equating coefficients

$$2x+1 = \underline{A}x - 5A + \underline{B}x + 7B$$

$$2x+1 = Ax+Bx - 5A+7B$$

$$2x+1 = (A+B)x + (-5A+7B)$$

coefficient of x

$$2 = A+B$$

$\searrow$
times 5

$$1 = -5A+7B$$

$$10 = 5A+5B$$

$$11 = 12B$$

$$\boxed{\frac{11}{12} = B}$$

$\leftarrow$ solve simultaneously

$$2 = A + \frac{11}{12}$$

$$\frac{24}{12} = A + \frac{11}{12}$$

$$\boxed{\frac{13}{12} = A}$$

Reminder: The denominator needs to have higher degree than numerator to do PF.

1.0.3 Example:

$$\int \frac{x^3 + 2x - 5}{x^2 + x} dx \quad \begin{array}{r} x-1 \\ x^2+x \overline{) x^3 - 5} \\ \underline{-(x^3 + x^2)} \\ -x^2 + 2x - 5 \\ \underline{-(-x^2 - x)} \\ 3x - 5 \end{array}$$

$$= \int x - 1 + \frac{3x-5}{x^2+x} dx$$

$$\left[\frac{3x-5}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1} \right] x(x+1)$$

↑ lower degree ∴
this is our remainder

$$3x-5 = A(x+1) + Bx$$

plug in $x = -1$

$$-3-5 = A(-1+1) + B(-1)$$

$$8 = B$$

plug in $x = 0$

$$-5 = A$$

$$= \int x - 1 - \frac{5}{x} + \frac{8}{x+1} dx = \boxed{\frac{1}{2}x^2 - x - 5\ln|x| + 8\ln|x+1| + C}$$

2 Case 2: Repeated linear factors in the denominator

2.0.1 Example:

$$\int \frac{2x+3}{(x+1)^3} dx \rightarrow \left[\frac{2x+3}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} \right] (x+1)^3$$

$$= 2x+3 = A(x+1)^2 + B(x+1) + C$$

plug in $x = -1$

use coefficients to get A+B.

$$2(-1)+3 = A(-1+1)^2 + B(-1+1) + C$$

$$\boxed{1 = C}$$

$$2x+3 = A(x^2+2x+1) + B(x+1) + C$$

coeff of x^2

$$0 = A$$

constant

$$3 = A + B + C$$

$$3 = 0 + B + 1$$

$$2 = B$$

$$\int \frac{0}{x+1} + \frac{2}{(x+1)^2} + \frac{1}{(x+1)^3} dx$$

$$u = x+1$$

$$du = dx$$

$$\frac{2}{u^2}$$

$$\frac{1}{u^3}$$

← Correction

$$\int 2u^{-2} + u^{-3} du$$

$$= \frac{2u^{-1}}{-1} + \frac{u^{-2}}{-2} + C = \boxed{-\frac{2}{x+1} - \frac{1}{2(x+1)^2} + C}$$